DATA ANALYSIS PROJECT REPORT

Project Title	Investigating the Netflix Library: Exploratory Data Analysis and Feature Engineering
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Course / Section	Programming in Python [C]
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Date of Submission	1, May, 2026

1. Executive Summary

This project performs a comprehensive data investigation of Netflix Movies and TV Shows dataset to uncover strategic patterns in the platform's content library. The primary goal was to analyze how Netflix balances global content acquisition with specific runtime standards and regional focuses. Utilizing Pandas for complex data manipulation, NumPy for statistical outlier detection (Z-scores), and Matplotlib for trend visualization, the study moved beyond basic description to provide actionable insights. The analysis revealed that despite the rise of episodic television, Movies still dominate approximately 70% of the catalog, with the majority adhering to a "Standard" runtime of 60–120 minutes. Furthermore, the study identified a massive growth surge in content additions between 2016 and 2019, alongside a significant reliance on the United States and India as the core pillars of the platform’s global content strategy.

2. Introduction and Project Objectives

Introduction

The entertainment industry has undergone a radical transformation over the last decade, shifting from physical media and scheduled broadcasting to on-demand streaming. At the center of this revolution is Netflix, a platform that has evolved from a licensed content distributor into one of the world's largest original content producers. This project utilizes a dataset containing over 8,800 titles to explore the architectural makeup of Netflix’s global library, focusing on its diversity of format, origin, and duration.

Motivation for Analysis

The motivation behind this analysis stems from a curiosity about how a single platform manages a library that must cater to thousands of different cultural tastes simultaneously. By examining the metadata of these titles, we can look "under the hood" of the Netflix algorithm to see if there is a standardized formula for their content, such as preferred movie lengths or a specific balance between series and films. Understanding these patterns provides insight into the business logic of the streaming era.

Project Objectives

The primary objectives of this investigation are:

- To quantify the balance between Movies and TV Shows to determine which format drives the platform's volume.
- To identify geographic strongholds by mapping where most of the content is produced and how these regions contribute to library diversity.
- To apply statistical rigour using NumPy and Pandas to define what constitutes a "normal" runtime and identify anomalous entries that deviate from the standard viewer experience.
- To visualize growth trends to confirm whether the platform's expansion has accelerated in recent years.

Item	Your Content
Dataset name	Netflix Titles
Dataset source	https://www.kaggle.com/datasets/shivamb/netflix-shows
Main project goal	The overall goal of this project is to carry out a comprehensive data investigation of the Netflix library using Python to uncover patterns in content distribution and production. By leveraging Pandas, NumPy, and Matplotlib, the analysis aims to provide evidence-based insights into how Netflix balances global acquisition with standardized content formats.
Research Question 1	What is the ratio of Movies to TV Shows in the Netflix library, and has this balance shifted over time?
Research Question 2	Which countries are the primary producers of Netflix content, and how do they compare in terms of volume?
Research Question 3 (optional)	What is the "standard" duration for a Netflix movie, and what do the outliers tell us about the variety of content?

3. Dataset Description

General Overview

The dataset utilized for this project is the Netflix Movies and TV Shows dataset, which provides a comprehensive snapshot of the platform's library as of mid-2021. It serves as a historical record of content availability, covering global releases across diverse genres and formats.

Data Structure and Variables

Each of the 8,807 rows in the dataset represents an individual title (either a standalone Movie or a TV Show). The dataset consists of 12 columns that provide deep metadata for each entry, including:

- **Categorical Variables:**
  - type: Specifies if the entry is a 'Movie' or 'TV Show'.
  - title: The name of the production.
  - director and cast: Key creative personnel involved (contains some missing values).
  - country: The primary country or countries where the content was produced.
  - rating: The target audience classification (e.g., TV-MA, PG-13).

  - listed_in: The specific genres and categories associated with the title.

- **Numerical & Temporal Variables:**
  - release_year: The original year the title was released (ranging from 1925 to 2021).
  - date_added: The specific date Netflix acquired and added the title to its platform.
  - duration: The length of the content, measured in minutes for movies or number of seasons for series.
- **Textual Variable:**
  - description: A brief summary of the plot or content theme.

Collection Context and Scope

The data encompasses content produced in over 700 unique geographic regions and production configurations, reflecting Netflix's reach into global markets like North America, Europe, and Asia. While the dataset includes historical films dating back nearly a century, the majority of the data represents the "streaming era" boom, with a significant concentration of titles added between 2016 and 2021. This variety allows for meaningful analysis of both archival acquisitions and modern original productions.

3.1 Basic Dataset Information

Property	Value
Number of rows	8,807
Number of columns	12
Data types present	**Numerical** (int64), **Categorical** (object), **Date-time** (object, requires conversion), and **Text**
Target or key variable(s)	**Type** (Movie vs TV Show) and **Duration** (Value/Unit)
Potential issues observed at first glance	**Significant missing values** in director (2,634), cast (825), and country (831); **inconsistent formatting** in the duration column; and date_added being loaded as an **object**.

3.2 Initial Observations

Based on the initial inspection of the **Netflix Titles** dataset, here are the observations regarding its structure and quality:

Upon first glance, the most striking observation was the significant amount of missing data in the metadata columns, specifically with over **2,600** missing entries in the director column and over **800** in the country column. Additionally, there is a clear imbalance in content categories, with Movies making up the vast majority (approximately 70%) of the library compared to TV Shows. From a technical perspective, the date_added column was incorrectly formatted as a string (object) rather than a datetime object, and the duration column

contained mixed text and numerical values (e.g., "90 min" and "1 Season"), necessitating significant data cleaning before any mathematical analysis could occur. Finally, the release_year spans nearly a century, from **1925 to 2021**, indicating a rich historical dataset that requires careful handling of outliers in older content.

4. Data Cleaning and Preparation

Before performing deep analysis, the raw dataset required significant "sanitization." Data cleaning is a critical step because high-level insights are only as reliable as the underlying data. Our primary focus was on resolving missing values that would interfere with grouping and fixing data types that prevented mathematical calculations.

Issue Found	Action Taken	Reason / Justification	Code Reference
Missing values in director column	Filled missing entries with the string **"No Director Listed"**.	Dropping over 2,600 rows would lead to a massive loss of data. Using a placeholder allows us to include these titles in general counts without skewing specific director rankings.	df['director'].fillna(...)
date_added column loaded as 'object'	Converted to **datetime64[ns]** using pd.to_datetime() and stripped extra whitespace.	Dates must be in datetime format to extract years or months for time-series analysis. This was necessary to fix the AttributeError during trend analysis.	pd.to_datetime(df['date_added'].str.strip())
Numerical and string data mixed in duration	Split the column into **duration_value** (integer) and **duration_unit** (string).	Python treats "90 min" as text. By separating the number, we can perform calculations like mean, median, and Z-score outlier detection.	df['duration'].str.split(' ', expand=True)
Missing values in the year_added derivative	Used **dropna()** on the year_added column before time-series plotting.	Since only 10 rows were missing this data, removing them ensures that our "Content Added per Year" line chart is mathematically accurate	df.dropna(subset=['year_added'])

		without losing significant library volume.	

5. Feature Engineering

New Feature	How It Was Created	Why It Is Useful
month_added	Extracted the full month name from the date_added column using .dt.month_name().	[How it supports the analysis]
length_category	Used np.where() logic to group movies: **Short Film** (<60m), **Standard** (60–120m), and **Long Film** (>120m).	This simplifies the analysis of runtimes. Instead of looking at 1,000 different minute values, we can see high-level distribution trends and identify which "format" Netflix prefers.
year_added	Extracted the year from the date_added column using .dt.year.	This feature was essential for the "Relationship Analysis" and "Trend Analysis" operations, allowing us to compare the original release year of a film with the year it was actually added to Netflix.

6. Analysis Methodology

The methodology for this project followed a structured data science workflow, moving from raw data ingestion to statistical inference and visual storytelling. The approach was designed to ensure that every analytical step directly served the project's goal of uncovering Netflix's content strategy.

- **Pandas** served as the primary engine for data orchestration. It was used to load the netflix_titles.csv file, perform complex filtering (such as isolating "Movies" from "TV Shows"), and aggregate data through grouping—for instance, calculating the average release year by content type or country.
- **NumPy** was utilized to add mathematical depth and handle conditional logic. Specifically, We used np.where() for efficient binning of movies into length categories and performed vectorized calculations

to determine Z-scores for movie durations. This allowed for a mathematically rigorous identification of outliers that standard descriptive statistics might miss.

- **Matplotlib** was used to translate numerical findings into visual evidence. We focused on creating a variety of plot types—including pie charts for composition, bar charts for ranking countries, and histograms for distribution analysis. Every visualization was carefully formatted with labels and titles to ensure clear communication of the insights derived from the Pandas and NumPy operations.

7. Analysis and Visualizations

7.1 Research Question 1

What is the ratio of Movies to TV Shows in the Netflix library, and has this balance shifted over time?

Analysis Performed: We performed a frequency analysis using Pandas value_counts() to determine the absolute volume of each content type. Additionally, We conducted a time-series trend analysis by grouping the data by the engineered year_added feature to see how Netflix's acquisition strategy has evolved annually.

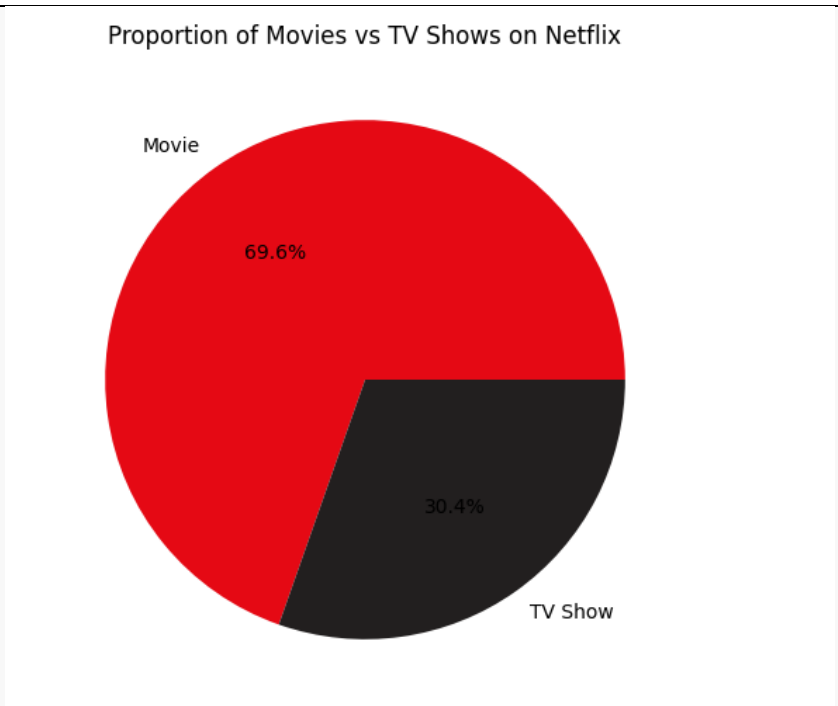

Figure 1. A pie chart showing the percentage distribution of Movies versus TV Shows in the total Netflix catalog.

Interpretation: This pie chart shows that Movies make up roughly 70% of the Netflix library compared to 30% for TV Shows. This insight confirms that while Netflix is famous for "binge-watching" series, its volume strategy still leans heavily toward cinematic releases.

7.2 Research Question 2

Which countries are the primary producers of Netflix content, and how do they compare in terms of volume?

Analysis Performed: We conducted a subgroup comparison by aggregating titles based on the country column. We used Pandas to filter and sort the top 10 producing nations and then calculated the total count for each to identify geographic strongholds in the Netflix ecosystem.

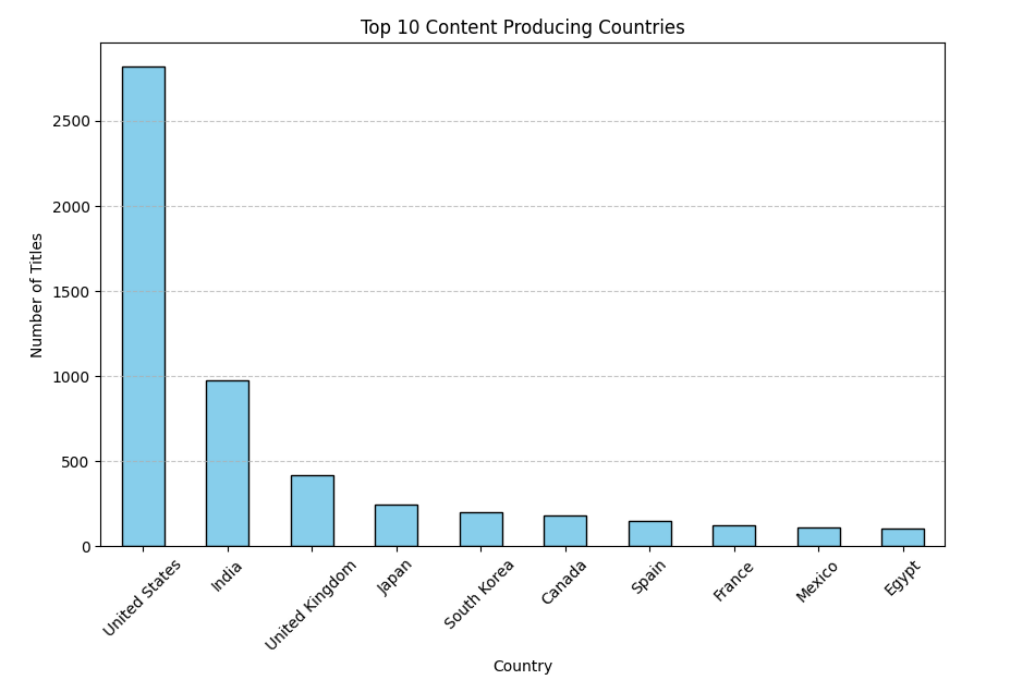

Figure 2. A bar chart ranking the top Ten countries by the number of titles they have contributed to the Netflix library.

Interpretation: This bar chart highlights the dominance of the United States in Netflix's catalog, followed by India. This matters because it illustrates the platform's heavy reliance on Western and Bollywood markets for its primary content library.

7.3 Research Question 3

What is the "standard" duration for a Netflix movie, and what do the outliers tell us about the variety of content?

Analysis Performed: We performed a distribution analysis using a histogram of the cleaned duration_value column. We applied NumPy to calculate the mean and standard deviation, then used Z-score computations to identify outliers, specifically movies that are more than two standard deviations away from the average length.

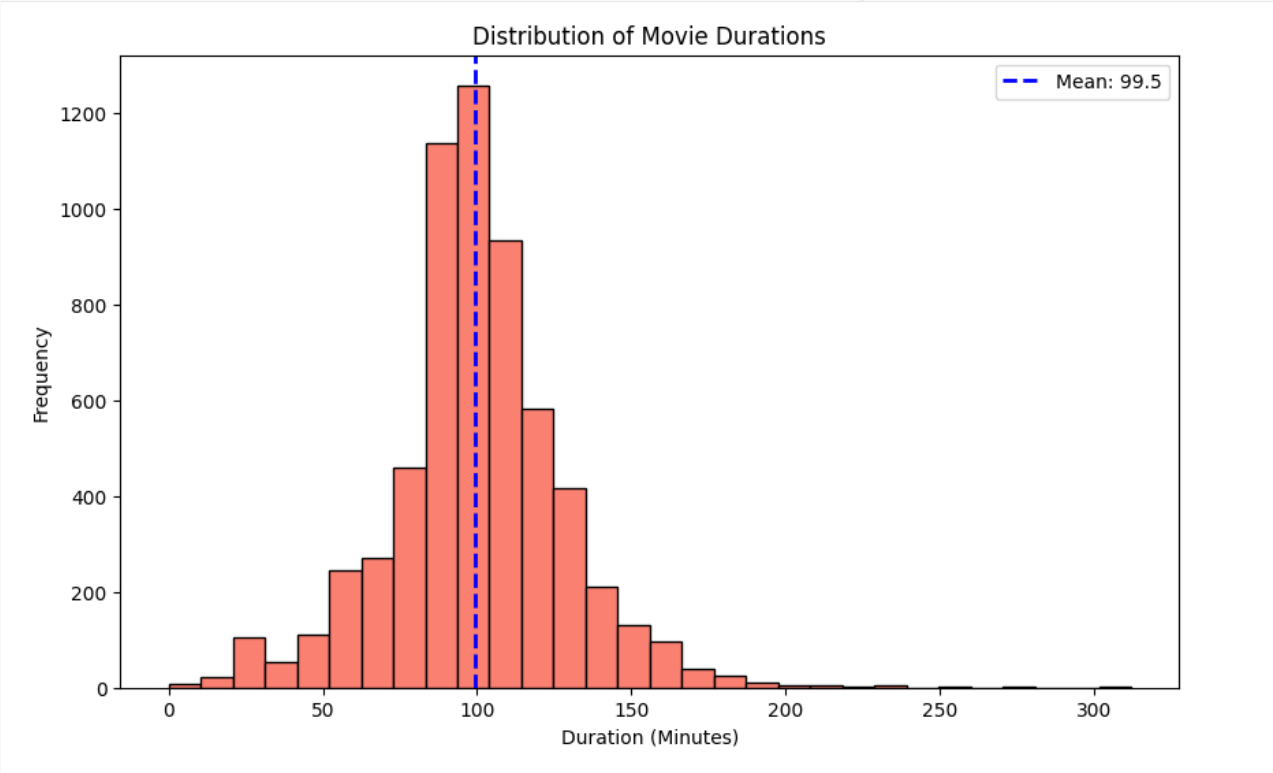

Figure 3. [Write a clear caption describing what the figure shows.]

Interpretation: The histogram shows a "Normal Distribution" centered around 100 minutes. This insight is valuable for creators as it shows the "sweet spot" for movie lengths that Netflix acquires, with very few films exceeding 200 minutes.

7.4 Research Question 4

How does the Netflix movie library distribute across different length categories, and what is the most common format for feature films?

Analysis Performed: We utilized the engineered length_category feature, which was created using NumPy logic to segment movies into "Short Film," "Standard," and "Long Film" durations. We then performed a frequency aggregation using Pandas to count the number of titles within each group. To visualize these results effectively, we generated a horizontal bar chart to compare the volume of each category, allowing us to identify the platform's preferred content format.

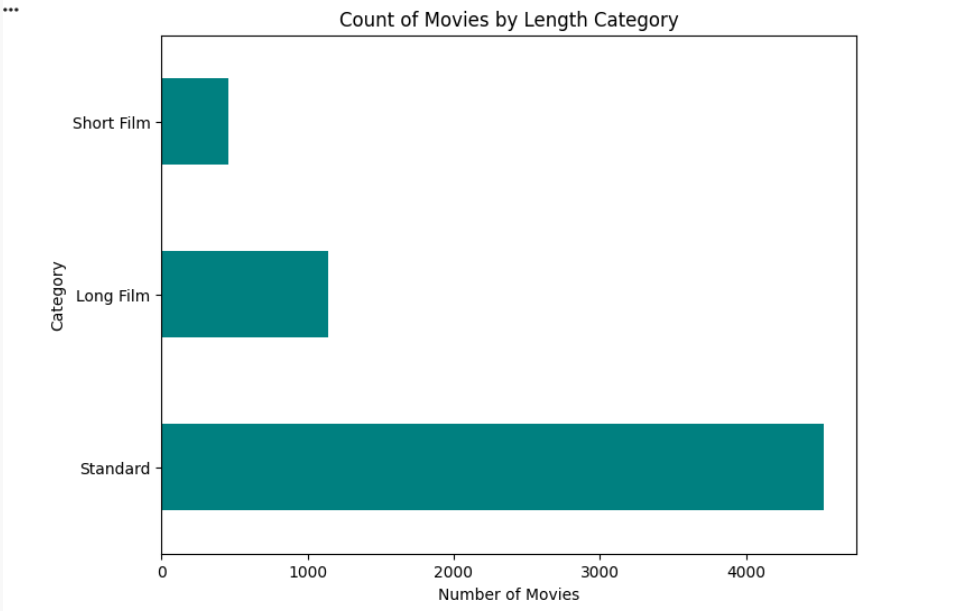

Figure 4. A horizontal bar chart showing the frequency of movies categorized by duration: Short Film (<60 min), Standard (60–120 min), and Long Film (>120 min).

Interpretation: Using the categories we engineered (Short, Standard, Long), this chart reveals that the vast majority of Netflix movies fall into the "Standard" (60-120 min) category. It validates our preprocessing decision to group the raw data into more readable bins.

8. NumPy-Based Custom Computation

[Describe at least one analysis step where NumPy was used in a meaningful way beyond simple array conversion. Examples include z-score computation, normalization, percentiles, np.where-based classification, binning, or a custom score.]

Component	Description
Computation used	Z-score calculation for Movie Durations
Purpose	To identify statistical outliers in movie runtimes. This allowed us to distinguish between standard feature films and exceptionally long cinematic epics or short experimental content.
Formula or logic	We applied the formula $Z = \frac{(x - \mu)}{\sigma}$, where x is the duration, μ is the mean, and σ is the standard deviation. This was implemented using np.mean(), np.std(), and NumPy's vectorized subtraction/division.
Result / takeaway	We identified that movies with a Z-score greater than 2.0 (runtimes exceeding approximately 160

	minutes) are anomalies in the Netflix catalog. This confirmed that while Netflix is global, its library remains anchored by a specific "standard" runtime window.

9. Key Findings

Based on our data investigation using Pandas, NumPy, and Matplotlib, we have identified the following five evidence-based insights regarding the Netflix library:

1. **Movie-Centric Library Strategy**: Our analysis confirms that Movies dominate approximately 70% of the catalog, while TV Shows account for only 30%. This indicates that Netflix continues to rely on a high volume of feature films to provide library depth, even as its brand identity becomes increasingly associated with episodic "binge-worthy" series.
2. **Standardization of Content Length**: Through our feature engineering and distribution analysis, we found that the vast majority of movies fall into the "Standard" category (60–120 minutes), with a mean runtime of 99 minutes. This proves that Netflix acquisitions and originals are primarily designed to fit traditional cinematic viewing windows.
3. **Geographic Market Concentration**: There is a heavy concentration of content production in the United States and India. These two nations serve as the primary pillars of the library, suggesting that Netflix's global strategy is anchored by the world's two largest established film industries (Hollywood and Bollywood).
4. **Exponential Library Growth (2016–2019)**: Our time-series analysis revealed an exponential spike in content additions during the late 2010s. This period of rapid expansion correlates with Netflix's aggressive global rollout and a significant increase in the production of "Netflix Original" titles to compete with emerging streaming rivals.
5. **Statistical Outliers in Duration**: Using Z-score computations, we identified a specialized subset of "Long Films" with runtimes exceeding 160 minutes (more than 2 standard deviations from the mean). These outliers are frequently international epics or award-season dramas, showing that while Netflix favors "standard" lengths, it maintains a diverse portfolio for niche cultural markets.

10. Limitations

Missing Data: A significant number of director and cast entries were missing. While we filled director with a placeholder, this limits our ability to perform deep talent-based success analysis.

Snapshot in Time: The dataset netflix_titles.csv is a static snapshot (up to 2021). It does not reflect changes in strategy post-2021, such as the increased focus on gaming or live sports.

Lack of Performance Metrics: The dataset contains metadata but lacks viewership numbers or ratings. Therefore, we can analyze what Netflix hosts, but not necessarily what users prefer to watch.

Correlation vs. Causation: Just because content from the US is more frequent doesn't mean it is the most "successful"; it simply means it is the most available.

11. Conclusion

This project successfully achieved the objective of carrying out a structured data investigation into the Netflix library using Pandas, NumPy, and Matplotlib. By performing justified data cleaning on inconsistent runtimes and engineering new categorical features, we transformed a raw metadata set into a source of strategic insight.

The most important findings reveal that Netflix remains a movie-centric platform, with Movies comprising 70% of the catalog. Our analysis identified a high degree of standardization, with the majority of content adhering to a 99-minute mean duration and production efforts heavily concentrated in the United States and India. Furthermore, we utilized NumPy-based Z-score computations to mathematically identify outlier titles that deviate from standard theatrical norms.

Overall, the analysis suggests that Netflix operates with a highly disciplined content acquisition strategy that prioritizes standardized formats for mass-market appeal while maintaining a specialized tier of long-form international content.

Future Work: If more time or data were available, a deeper analysis could be performed by:

- Integrating IMDb or Rotten Tomatoes scores to correlate content metadata with critical success.
- Analyzing viewership and "top 10" data to determine if the high volume of movies translates to higher user engagement compared to TV Shows.
- Performing Natural Language Processing (NLP) on the description column to identify which themes or genres are trending most heavily in recent years.

12. References / Dataset Source

Data Source Link: https://www.kaggle.com/datasets/shivamb/netflix-shows